Test Plan (QA) — Offline Bangla Dubbing Video Player

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QA verifies the version 1 build against BRD v1.1 in a 3-week test cycle, then hands a release candidate to step 6, UAT.

Entry criteria

These match the Development Plan's handoff checklist.

- [ ] All Must requirements are done and merged to `main`
- [ ] Installers for Windows, macOS and Linux built from a tagged release candidate
- [ ] Release notes list features, known issues and supported hardware
- [ ] Test fixtures and expected results are in `tests/fixtures`
- [ ] Open bugs: none critical, at most 5 major
- [ ] Nightly GPU and no-network CI checks green for the last 3 nights

Scope and approach

In scope: every BRD v1.1 business requirement (BR-01 to BR-17, except BR-04b) and non-functional requirement (NFR-01 to NFR-10), on all three platforms.

Out of scope: BR-04b Bangladeshi accent (phase 2), lip-sync, streaming sources, mobile.

Test type	What it checks	How
Functional	Each requirement's acceptance condition	Manual test cases below
Regression	Nothing broke between builds	Automated suite from development (unit, UI, integration) re-run on each build
Quality	Translation, voice and sync as Bangladeshi viewers hear them	Reviewer scoring protocol
Performance	Speed, live-mode buffer, memory	Timed runs on reference machines

Test type	What it checks	How
Offline and privacy	No network after setup; no data leaves the device	Network blocked and traffic monitored
Compatibility	OS versions, GPU / CPU, video formats	Test matrix
Installation	Install, first-run model download, upgrade, uninstall	Clean machines
Robustness	Fallbacks and error handling	Fault injection: bad files, no GPU memory, killed worker

Test environments and test data

Reference machines

ID	Machine	OS	Role
M1	Desktop, NVIDIA RTX GPU 8 GB, 16 GB RAM	Windows 11	Recommended hardware
M2	Desktop, NVIDIA GPU 8 GB+	Ubuntu 22.04	Recommended hardware, Linux
M3	MacBook, Apple Silicon, 16 GB	macOS 14	Recommended hardware, macOS
M4	Laptop, CPU only, 8 GB RAM	Windows 10	Minimum hardware
M5	Mac, macOS 12	macOS 12	Oldest supported macOS

Test videos

ID	Content	Length	Tests
V1	English, 1 speaker, lecture	5 min	Baseline
V2	English, 4 speakers, drama	5 min	BR-05, BR-17
V3	Hindi, 2 speakers	5 min	BR-02 guaranteed language

ID	Content	Length	Tests
V4	Korean, 2 speakers	5 min	Best-effort pivot
V5	English, loud music and effects	5 min	BR-06 and fallback
V6	English feature film	2 h	Speed, live mode, stability
V7	English, 7 speakers	10 min	Over-maximum speaker handling
V8	Corrupt / audio-only / no-speech files	Short	Robustness
V9	Same content in MP4, MKV, AVI, MOV, WebM	1 min each	Formats

Only videos the team owns or has rights to test are used. Each has a reference transcript and speaker list for scoring.

Functional test cases

Run on M1, M2 and M3 unless noted. Record Pass / Fail and the defect ID in the Result column.

ID	Req	Steps	Expected result	Result
TC-01	BR-01	Open V1; use play, pause, seek, volume, fullscreen	All controls respond within 0.5 s; seek lands within 1 s of target	
TC-02	BR-02	Dub V1, V3, V4 with auto-detect	Detects en, hi, ko; V4 labelled "best-effort"	
TC-03	BR-02	Dub V3 with language forced to English	Override applied; output reflects the forced language	
TC-04	BR-03	Dub V1; play with Bangla audio	Every speech segment of 1 s+ is spoken in Bangla; no subtitles needed	
TC-05	BR-04a	Dub V1 and V2; compare output with the glossary	Glossary terms used in 95%+ of matching lines	

ID	Req	Steps	Expected result	Result
TC-06	BR-05	Dub V2	4 distinct voices; 90%+ lines on the correct speaker; each speaker keeps one voice	
TC-07	BR-05	Dub V7 (7 speakers)	Smallest speakers merged into 6 voices; user is told	
TC-08	BR-06	Dub V5	Music and effects audible under Bangla voices	
TC-09	BR-06	Dub V5 with separation forced to low quality	Fallback: original audio lowered, dub on top; no crash	
TC-10	BR-07	Disconnect network; dub V1	Completes; no network attempts (see NFR-03 test)	
TC-11	BR-08	Dub V6 in pre-process mode, then play	Plays start to end in sync; no stalls	
TC-12	BR-09	Start V6 in live mode on M1	Playback starts within 60 s; no buffering pauses for 2 h	
TC-13	BR-09	Open settings on M4	Live mode hidden or disabled with an explanation	
TC-14	BR-10	Change Speaker 2′s voice; re-dub	Only Speaker 2′s lines change voice	
TC-15	BR-11	Toggle audio Bangla / Original during playback	Switches within 1 s without losing position	
TC-16	BR-12	Export V1 as WAV and as MKV	Files open in VLC; MKV has Bangla and Original tracks with labels	
TC-17	BR-13	Turn on Bangla subtitles	Subtitles match the spoken Bangla and timing	
TC-18	BR-14	Re-open a dubbed video; rename it and re-open	Plays dub immediately both times, no reprocessing	
TC-19	BR-15	Edit a Bangla line; regenerate	New audio replaces only that line within 30 s	

ID	Req	Steps	Expected result	Result
TC-20	BR-16	Switch UI language to Bangla	All menus, dialogs and errors in Bangla; no clipped text	
TC-21	BR-17	Merge two speakers in V2	Merged lines use one voice; other lines unchanged	
TC-22	BR-17	Split a speaker; rename both	Re-dub affects only the split lines; names persist after restart	
TC-23	NFR-06	Kill the worker process mid-dub	UI restarts it and resumes from the last finished stage	
TC-24	NFR-06	Open V8 files	Clear error message for each; app stays open	
TC-25	—	Open each V9 format	All play and dub	

Non-functional tests

ID	Req	Test	Pass condition	Result
NT-01	NFR-01	Install and dub V1 on Windows 10, 11; macOS 12, 14; Ubuntu 22.04, 24.04	Works on all	
NT-02	NFR-02	Dub V1 on M4 (CPU only)	Completes in pre-process mode; live mode disabled	
NT-03	NFR-03	Dub V1–V5 with a traffic monitor (Wireshark or Little Snitch) running	Zero outbound connections after model download	
NT-04	NFR-03	Inspect logs after dubbing	No transcripts, translations or file contents in logs	
NT-05	NFR-04	First launch on a clean machine; cut network at 50% of download; reconnect	Download resumes; checksums pass; total size 5–10 GB shown before starting	

ID	Req	Test	Pass condition	Result
NT-06	NFR-05	Measure start offset of every line in V1, V2, V3 against the reference	95% within 0.5 s; none over 1.0 s	
NT-07	NFR-06	Force one line to fail in TTS	That line plays original audio; playback continues	
NT-08	NFR-07	5 first-time users dub V1 without help	Each finishes in 3 clicks or fewer	
NT-09	NFR-08	Compare shipped models and libraries with the licence list	Only commercial-safe licences; NLLB-200 and MMS-TTS absent	
NT-10	NFR-09	Install, upgrade from previous build, uninstall on each OS	Clean install and removal; cache kept on upgrade	
NT-11	NFR-10	Pre-process V6 on M1, M2, M3	Under 15 min per hour of video	
NT-12	—	Dub V6 three times in a row on M1	No memory growth over 10%; no crash	

All times and offsets are recorded in the test report, per machine.

Quality scoring protocol

Translation and voice quality can only be judged by listeners, so QA runs a structured review on V1–V5.

Reviewers: at least 5 native Bangladeshi Bangla speakers, mixing ages and regions; none from the development team.

Method:

1. Each reviewer watches the 5 dubbed clips in random order, without the original audio first.
2. They score each clip from 1 (poor) to 5 (excellent) on the criteria below.
3. They then watch with original audio available and note specific bad lines (time and issue).
4. QA averages the scores and turns noted lines into defects or glossary additions.

Criterion	Question for reviewers	Pass (average)
Understanding	Could you follow what was said without subtitles?	4.0
Translation	Is the Bangla correct and natural?	3.5
Bangladeshi wording	Does the vocabulary sound Bangladeshi? (BR-04a)	3.5
Speaker voices	Could you tell the characters apart?	4.0
Timing	Did voices match when people spoke?	3.5
Background	Was music and sound still natural?	3.5
Accent (informational)	Does the voice sound Bangladeshi?	Recorded for phase 2

Scores below pass on any criterion are logged as a major defect against the related requirement.

Defect management

Severity	Definition	Example	Fix target
Critical	Crash, data loss, network use offline, or a Must requirement fails	App crashes on MKV; video sent online	Before next build
Major	Should requirement fails, or a quality score below pass	Wrong speaker on 20% of lines; sync off by 2 s	Within the test cycle
Minor	Works with a workaround or small visual issue	Bangla label clipped in settings	Before release if time allows
Trivial	Cosmetic, no user impact	Icon misaligned by 2 px	Backlog

Each defect records: build version, machine ID, OS, test case ID, test video and timestamp, steps, expected vs actual, logs from "Copy diagnostics".

Workflow:

1. QA logs the defect as **New** with severity.
2. Daily triage (QA + lead developer + product owner) sets priority and owner, or rejects it.
3. Developer fixes on a `fix/` branch → **Fixed** in the next build.
4. QA re-tests the case and its neighbours → **Closed** or **Reopened**.
5. Defects that trace to a requirement gap go to the product owner for a BRD change, not a code fix.

Schedule, reporting and exit criteria

Week	Activity
1	Installation, compatibility, functional tests TC-01 to TC-25 on all machines
2	Non-functional tests, quality review sessions, fault injection; first fix build
3	Re-test fixes, full regression on the release candidate, final report

Reporting: a daily status (cases run, passed, failed, open defects by severity) in the team channel, and a final test report with every result, timing and quality score.

Exit criteria to step 6, UAT:

- [ ] 100% of test cases run; 95%+ pass, and every Must-requirement case passes
- [ ] Zero open critical defects; at most 3 open major defects, each with a known workaround
- [ ] Quality scores at or above pass on every criterion
- [ ] Offline and privacy tests (NT-03, NT-04) pass on all three platforms
- [ ] Speed (NT-11) and sync (NT-06) pass on recommended hardware
- [ ] Installers signed and tested on clean machines
- [ ] Final test report approved by the product owner

If exit criteria are not met: another fix cycle of up to 1 week. If a failure comes from a model limit rather than a bug, the product owner applies the BRD v1.1 fallback (for example, moving live mode to version 2) and QA re-tests the reduced scope.

Change log

Version	Date	Change
1.0	2026-10-29	First version, based on BRD v1.1 and Development Plan v1.1